

Why every triangle adds up to 180

Where proof begins

The big idea. Tear the three corners off any triangle and they snap together into a straight line. A straight line is 180° , so the three angles add to 180. The real question: *why must this happen for every triangle, even ones nobody has drawn yet?*

Activity: tear three different triangles

1. Draw three triangles that look nothing alike: a long skinny one, a fat one, a pointy one.
2. Shade the three corners of each so you can tell them apart.
3. **Tear** (do not cut neatly) the three corners off each triangle.
4. Slide the three torn corners together, points touching, and see what they make.

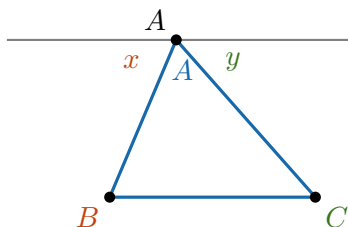
Draw your three triangles here, then tape the reassembled corners below them:



Every time, the three corners make a _____, which is _____ degrees.

The proof: a parallel line through the top

Take any triangle ABC . Through the top vertex A , draw a line *parallel to the bottom side* BC . Three angles sit along that straight line at A : x , the apex angle A , and y .



Because x , A , y lie along a straight line, $x + A + y = 180$. The “Z” (alternate-angles) rule says $x = B$ and $y = C$, because of the parallel lines.

Substitute $x = B$ and $y = C$ into $x + A + y = 180$. What do you get? Why does this prove it for *every* triangle and not just yours?

Push further

1. **Four sides.** Split any quadrilateral into two triangles with one diagonal. Its four angles add to _____ degrees.
2. **Many sides.** An n -sided polygon splits into $n - 2$ triangles, so its angles add to $(n - 2) \times 180$. Check it on a pentagon (5 sides $\Rightarrow$ _____).
3. **Competition gem: the five-pointed star.** Draw a five-pointed star in one stroke. **What do the five point-angles add up to?** (Hint: each point is the tip of a little triangle; chase the angles with the 180 rule.)

4. **True or false?** Guess first, then check.

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| • The three angles of a triangle always add to 180. | T / F |
| • A very pointy triangle adds to <i>more</i> than 180. | T / F |
| • The five points of <i>any</i> star add to 180. | T / F |